

Lecture 04 - Diffusion Models Today: Score SDE Framework

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Topics on Diffusion Models

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Introduction: Ordinary Differential Equations (ODEs)

- **Core Definition:** ODEs describe how a system's state changes over time according to a precise rule - knowing the starting point determines the future path exactly.

$$\frac{dx(t)}{dt} = v(x(t), t)$$

- **Geometric Interpretation:** Solving the ODE means tracing a curve $x(t)$ where the tangent (instantaneous velocity) at any point aligns with the vector $v(x(t), t)$.

Introduction: Stochastic Differential Equations (SDEs) - Discrete Form

- **Motivation:** ODEs ignore randomness, but real-world systems are affected by noise/uncertainty. SDEs extend ODEs to incorporate stochasticity.
- **Discrete Update Rule (From ODE to SDE):** Augment ODE with a stochastic term to model random perturbations:

$$x_{t+\Delta t} = x_t + f(x_t, t)\Delta t + g(t)\sqrt{\Delta t} \cdot \varepsilon_t, \quad (\text{A.2.2})$$

- $\varepsilon_t \sim \mathcal{N}(0, I_D)$: i.i.d. standard Gaussian vectors;
- $g : [0, T] \rightarrow \mathbb{R}$: diffusion coefficient (time-dependent only here).

Introduction: SDEs - Continuous Form & Wiener Process

- **Continuous SDE (Limit of Discrete Form):** As $\Delta t \rightarrow 0$, Equation (A.2.2) defines the continuous SDE:

$$dx(t) = f(x(t), t)dt + g(t)dw(t), \quad (\text{A.2.3})$$

where $w(t) \in \mathbb{R}^D$ is a **Wiener process** (standard Brownian motion).

- **Key Properties of Wiener Process:**
 - Initial State: $w(0) = 0$ almost surely;
 - Independent Increments: For $0 \leq s < t$, $w(t) - w(s)$ is independent of the past;
 - Gaussian Increments: $w(t) - w(s) \sim \mathcal{N}(0, (t - s)I_D)$ (A.2.4);
 - Continuity: Sample paths $t \mapsto w(t)$ are continuous (a.s.) but nowhere differentiable.

From DDPM to Forward-Time SDEs

- **DDPM Noise Injection:** DDPM injects noise incrementally with a variance schedule $\{\beta_i\}_{i=1}^L$:

$$x_i = \sqrt{1 - \beta_i} x_{i-1} + \sqrt{\beta_i} \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, I)$$

- **Continuous Approximation of DDPM:**

$$x_{t+\Delta t} = \sqrt{1 - \beta_t} x_t + \sqrt{\beta_t} \varepsilon_t \approx x_t - \frac{1}{2} \beta_t x_t \Delta t + \sqrt{\beta_t \Delta t} \varepsilon_t$$

- **Unification to Continuous-Time SDE:** Discrete methods (NCSN, DDPM) can be unified via a forward SDE on $[0, T]$:

$$dx(t) = f(x(t), t)dt + g(t)dw(t), \quad x(0) \sim p_{\text{data}}$$

where $f(\cdot, t) : \mathbb{R}^D \rightarrow \mathbb{R}^D$ (drift), $g(t) \in \mathbb{R}$ (diffusion coefficient), $w(t)$ (Wiener process).

SDE Closed-Form Solution: Mean $m(t)$

- **Motivation:** Derive mean-variance ODEs to analyze diffusion term's impact on final variance with affine transformation where $f(x, t) = f(t)x$.
- **Mean $m(t) = \mathbb{E}[x(t)|x_0]$ Derivation:**

$$\begin{aligned}\frac{dm(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}[x(t + \Delta t)|x_0] - \mathbb{E}[x(t)|x_0]}{\Delta t} \\ x(t + \Delta t) &\approx x(t) + f(t)x(t)\Delta t + g(t)\Delta w(t) \\ \frac{dm}{m} &= f(t)dt \\ \ln\left(\frac{m(t)}{m(0)}\right) &= \int_0^t f(u)du \\ m(t) &= \exp\left(\int_0^t f(u)du\right)x_0\end{aligned}$$

SDE Closed-Form Solution: Covariance $P(t)$

- **Covariance Definition:** $P(t) = \text{Var}(x(t)) = \mathbb{E}[\Delta x(t)\Delta x(t)^\top]$, where $\Delta x(t) = x(t) - m(t)$.
- **Differential of $\Delta x(t)$:**

$$\begin{aligned}
 d\Delta x(t) &= dx(t) - dm(t) \\
 &= [f(t)x(t)dt + g(t)dw(t)] - [f(t)m(t)dt] \\
 &= f(t) \underbrace{(x(t) - m(t))}_{\Delta x(t)} dt + g(t)dw(t) \\
 d\Delta x(t) &= f(t)\Delta x(t)dt + g(t)dw(t)
 \end{aligned}$$

SDE Closed-Form Solution: Covariance $P(t)$

- Differential of $\Delta x(t)$:

$$\Delta x(t) = f(t)\Delta x(t)dt + g(t)dw(t)$$

- **Key Insight on Variance:** The variance change is modulated by drift (contraction/expansion) + diffusion (noise accumulation), not just diffusion alone.
- **Covariance Closed-Form:**

$$P(t) = \int_0^t \exp \left(2 \int_s^t f(u)du \right) g^2(s)ds$$

Probability Flow Foundation: Probability Density Evolution

- **Definition of Probability Flow:** Define $j_t(x) = v_t(x)p_t(x)$ (probability flux: probability passing through unit area per unit time):

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot j_t(x)$$

Interpretation: The time change rate of density equals the negative divergence of probability flow.

- **Core Physical Essence:** Describe the intrinsic connection between the evolution of probability density and the sample velocity field, starting from the **law of probability conservation**.

Probability Flow Foundation: Probability Density Evolution

- **Basic Evolution Equation of Probability Density:** If samples are driven by a velocity field $v_t(x)$, the time change rate of $p_t(x)$ is:

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot [v_t(x)p_t(x)]$$

where $\nabla_x \cdot$ denotes the spatial divergence operator, and the negative sign reflects probability conservation.

- **Two-Factor Decomposition (Divergence Product Rule):** Apply $\nabla_x \cdot (AB) = (\nabla_x \cdot A)B + A \cdot \nabla_x B$:

$$\frac{\partial p_t(x)}{\partial t} = -(\nabla_x \cdot v_t(x)) p_t(x) - v_t(x) \cdot \nabla_x p_t(x)$$

Two core driving factors for density evolution: geometric properties of velocity field (divergence) + inhomogeneity of density distribution (gradient).

Probability Flow Foundation: Evolution Mechanism & Definition

- Factor 1: Geometric Property of Velocity Field (Divergence Impact):** Term $-(\nabla_x \cdot v_t(x)) p_t(x)$ modulates density via "divergence-convergence" of velocity field:
 - $\nabla_x \cdot v_t(x) > 0$ (divergence): Samples spread outward, density decreases;
 - $\nabla_x \cdot v_t(x) < 0$ (convergence): Samples gather locally, density increases;
 - $\nabla_x \cdot v_t(x) = 0$ (irrotational field): No divergence/convergence, only Factor 2 works.
- Factor 2: Inhomogeneity of Density Distribution (Gradient Impact):** Term $-v_t(x) \cdot \nabla_x p_t(x)$ reflects the coupling effect of density gradient and velocity field:
 - $\nabla_x p_t(x)$ points to the direction of increasing density;
 - Inner product $v_t \cdot \nabla_x p_t$ describes the transport of density differences (inverse gradient direction $\rightarrow$ density increases in low-density regions).

- **Forward Stochastic Differential Equation (SDE):** The forward noising process in diffusion models is described by:

$$dx(t) = f(x(t), t)dt + g(t)dw(t)$$

where:

- $f(x(t), t)$: Drift term (deterministic velocity field);
 - $g(t)$: Diffusion coefficient (intensity of random noise);
 - $dw(t)$: Standard Wiener process increment, $\sim \mathcal{N}(0, dt)$.
- **Fokker-Planck Equation (FPE):** The probability density evolution equation corresponding to forward SDE is FPE:

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot [f(x, t)p_t(x)] + \frac{1}{2}\nabla_x^2 [g^2(t)p_t(x)]$$

Diffusion Process Extension: Forward SDE & Fokker-Planck Equation

- **Derivation Link to Probability Flow:** From the basic probability conservation equation (Eq. (3)): $\frac{\partial p_t}{\partial t} = -\nabla_x \cdot j_t(x)$. We need to rearrange FPE into this "negative divergence of probability flow" form to derive $j_t(x)$.
- **Step 1: Expand the Laplacian Term in FPE** Apply the product rule for gradient: $\nabla_x(AB) = (\nabla_x A)B + A\nabla_x B$. Since $g^2(t)$ is time-dependent only (independent of x), $\nabla_x g^2(t) = 0$. Thus:

$$\begin{aligned}\nabla_x^2 [g^2(t)p_t(x)] &= \nabla_x \cdot [\nabla_x (g^2(t)p_t(x))] \\ &= \nabla_x \cdot [g^2(t)\nabla_x p_t(x) + p_t(x)\nabla_x g^2(t)] \\ &= \nabla_x \cdot [g^2(t)\nabla_x p_t(x)]\end{aligned}$$

Diffusion Process Extension: Derivation of Probability Flow Decomposition

- Step 2: Rearrange FPE into Negative Divergence Form Substitute the expanded Laplacian term back into FPE (Eq. (5)):

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot [f(x, t)p_t(x)] + \frac{1}{2}\nabla_x \cdot [g^2(t)\nabla_x p_t(x)]$$

Factor out the divergence operator (linearity of $\nabla_x \cdot$):

$$\frac{\partial p_t(x)}{\partial t} = -\nabla_x \cdot \left[f(x, t)p_t(x) - \frac{1}{2}g^2(t)\nabla_x p_t(x) \right]$$

Diffusion Process Extension: Derivation of Probability Flow Decomposition

- Step 3: Match Basic Probability Conservation Equation Compare with Eq. (3): $\frac{\partial p_t}{\partial t} = -\nabla_x \cdot j_t(x)$. The term inside the divergence is exactly the probability flow $j_t(x)$:

$$j_t(x) = \underbrace{f(x, t)p_t(x)}_{\text{Drift Flow } j_{\text{drift}}} + \underbrace{-\frac{1}{2}g^2(t)\nabla_x p_t(x)}_{\text{Diffusion Flow } j_{\text{diffuse}}}$$

Diffusion Process Extension: Physical Meaning of Probability Flow Decomposition

- **Drift Flow** $\mathbf{j}_{\text{drift}} = \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})$: Directed transport driven by the deterministic drift term $\mathbf{f}(\mathbf{x}, t)$. It reflects the "ordered movement" of samples along the drift direction, and its magnitude is proportional to both the drift intensity and the local probability density.
- **Diffusion Flow** $\mathbf{j}_{\text{diffuse}} = -\frac{1}{2}g^2(t)\nabla_{\mathbf{x}}p_t(\mathbf{x})$: Irregular transport induced by random noise. The key properties are:
 - Negative sign: Points to the direction opposite to the density gradient (from high-density to low-density regions);
 - Proportional to $g^2(t)$: The stronger the noise intensity, the more significant the diffusion effect;
 - Proportional to $\nabla_{\mathbf{x}}p_t(\mathbf{x})$: The greater the density inhomogeneity, the stronger the diffusion flux.

Physical Basis of Diffusion Flow: Fick's Law

- **Motivation:** The form of diffusion flow in FPE is not arbitrary, but derived from **Fick's Law** (classical law for molecular diffusion).
- **Basic Form of Fick's Law:** Core conclusion: Diffusion flux is proportional to concentration gradient and opposite in direction (from high to low density). Replace "concentration" with "probability density":

$$j_{\text{diffuse}}(x) = -D(t)\nabla_x p_t(x)$$

where:

- $D(t)$: Diffusion coefficient (reflects the speed of diffusion process);
- Negative sign: Diffusion flux points to the opposite direction of density gradient (inverse gradient diffusion).

Physical Basis of Diffusion Flow: Fick's Law

- **Link Between Fick's Law and FPE Diffusion Flow:** Compare Eq. (7) with the diffusion flow term in Eq. (6):

$$-\frac{1}{2}g^2(t)\nabla_x p_t(x) = -D(t)\nabla_x p_t(x)$$

Directly derive the relationship between diffusion coefficient and SDE diffusion term:

$$D(t) = \frac{1}{2}g^2(t)$$

Interpretation: Diffusion flow in FPE is the direct application of Fick's Law in probabilistic diffusion scenarios.

Validation of Key Coefficient ($\frac{1}{2}$): $It\hat{o}$ Integral & Einstein Relation

- **Mathematical Source:** $It\hat{o}$ Integral Correction Forward SDE is a typical $It\hat{o}$ process, whose integral rule differs from ordinary calculus (non-zero high-order terms):

$$d(x(t)^2) = 2x(t)dx(t) + (dx(t))^2$$

$$(dx(t))^2 = g^2(t)dw(t)^2, \quad \mathbb{E} \left[(dx(t))^2 \right] = g^2(t)dt$$

This high-order term directly contributes to the $\frac{1}{2}$ coefficient in FPE/diffusion flow.

Validation of Key Coefficient ($\frac{1}{2}$): It $\hat{t}$ Integral & Einstein Relation

- **Physical Source: Einstein Relation** Einstein established the link between diffusion coefficient and thermal motion velocity (Einstein Relation):

$$D = \frac{1}{2} \langle v^2 \rangle \tau$$

where:

- $\langle v^2 \rangle$: Velocity variance (proportional to $g(t)^2$ in SDE);
- τ : Particle collision time.

This verifies that $D \propto g^2(t)$ with proportionality coefficient $\frac{1}{2}$ (consistent with Eq. (8)).

Introduction of Score Function: Definition & Velocity Field Simplification

- **Motivation:** Simplify the diffusion flow expression and clarify its physical meaning by defining the "gradient of unit probability density".
- **Equivalent Velocity Field from Probability Flow** From $j_t(x) = p_t(x)v_t(x)$ and Eq. (6), solve for the equivalent velocity field:

$$v_t(x) = \frac{j_t(x)}{p_t(x)} = f(x, t) - \frac{1}{2}g^2(t) \cdot \frac{\nabla_x p_t(x)}{p_t(x)}$$

Introduction of Score Function: Definition & Velocity Field Simplification

- **Definition of Score Function** Define the "gradient of log probability density" (score function) as the gradient of unit density:

$$\nabla_x \log p_t(x) = \frac{\nabla_x p_t(x)}{p_t(x)}$$

Physical meaning: Points to the direction of fastest density growth, magnitude reflects growth rate per unit density.

- **Velocity Field in Score Function Form** Substitute Eq. (10) into Eq. (9) to simplify the velocity field:

$$v_t(x) = f(x, t) - \frac{1}{2}g^2(t)\nabla_x \log p_t(x)$$

Construction of Probability Flow ODE (PF-ODE): Derivation & Core Properties

- **Derivation of PF-ODE** The time derivative of the sample trajectory equals the equivalent velocity field. Let $\tilde{x}(t)$ denote the PF-ODE trajectory:

$$\frac{d\tilde{x}(t)}{dt} = v_t(\tilde{x}(t))$$

Substitute Eq. (11) (score function velocity field) to get the final PF-ODE:

$$\frac{d\tilde{x}(t)}{dt} = f(\tilde{x}(t), t) - \frac{1}{2}g^2(t)\nabla_x \log p_t(\tilde{x}(t))$$

Construction of Probability Flow ODE (PF-ODE): Derivation & Core Properties

● Core Properties of PF-ODE

- **Deterministic:** No random term $dw(t)$, same initial condition $\rightarrow$ same trajectory (efficient and stable numerical integration);
- **Distribution Matching:** Trajectory $\tilde{x}(t)$ follows the marginal distribution $p_t(x)$ of forward SDE, ensuring final samples match real data distribution;
- **Bidirectional:** Forward integration ($t = 0 \rightarrow T$): real data $\rightarrow$ Gaussian noise; Reverse integration ($t = T \rightarrow 0$): Gaussian noise $\rightarrow$ real data (generation).
- **Logical Closure:** PF-ODE unifies "distribution matching" (from probability flow) and "efficient generation" (deterministic ODE), core of diffusion-based generative models.

Derivation Overview: Reverse SDE via Bayes & Taylor Expansion

- **Core Goal:** Derive the reverse-time SDE (denoising process) from the forward SDE (noising process), with the score function naturally introduced.
- **Key Tools:**
 - Bayes' Theorem: Invert forward conditional probability to reverse conditional probability;
 - Taylor Expansion: Approximate small perturbations (valid for $\Delta t \rightarrow 0$);
 - Gaussian Distribution Property: Forward/reverse transfer probabilities are both Gaussian.

Derivation Premise: Forward SDE & Transfer Probability

- Forward SDE (Noising Process):

$$dx(t) = f(x(t), t)dt + g(t)dw(t)$$

where:

- $f(x, t)$: Forward drift term;
- $g(t)$: Scalar diffusion coefficient (x -independent);
- $dw(t) \sim \mathcal{N}(0, \Delta t \cdot I)$ (Wiener increment).

Derivation Premise: Forward SDE & Transfer Probability

- **Forward Conditional Probability:** Due to Gaussianity of Brownian motion, the transfer probability for $\Delta t \rightarrow 0$ is Gaussian:

$$p(x_{t+\Delta t} | x_t) = \mathcal{N}(x_{t+\Delta t} | \mu_{\text{forward}}, \Sigma_{\text{forward}})$$

- Mean: $\mu_{\text{forward}} = x_t + f(x_t, t)\Delta t$;
- Covariance: $\Sigma_{\text{forward}} = g^2(t)\Delta t \cdot I$.
- **Logarithmic Form (Key for Derivation):**

$$\log p(x_{t+\Delta t} | x_t) \approx -\frac{1}{2g^2(t)\Delta t} \|x_{t+\Delta t} - x_t - f(x_t, t)\Delta t\|^2 + C$$

(C: constant term, negligible in derivation)

Core Step 1: Invert via Bayes' Theorem

- **Target:** Get reverse conditional probability $p(x_t | x_{t+\Delta t})$ (denoising direction).
- **Bayes' Theorem:**

$$p(x_t | x_{t+\Delta t}) = \frac{p(x_{t+\Delta t} | x_t) \cdot p(x_t)}{p(x_{t+\Delta t})}$$

- **Logarithmic Transformation** (monotonic function, preserves distribution shape):

$$\log p(x_t | x_{t+\Delta t}) = \log p(x_{t+\Delta t} | x_t) + \log p(x_t) - \log p(x_{t+\Delta t}) + C$$

- **Rearrange Key Term:** Substitute Eq. (3) into Eq. (5), ignore constant C :

$$\log p(x_t | x_{t+\Delta t}) \approx -\frac{1}{2g^2(t)\Delta t} \|\Delta x - f\Delta t\|^2 - (\log p(x_{t+\Delta t}) - \log p(x_t))$$

where $\Delta x = x_{t+\Delta t} - x_t$ (small perturbation, $\Delta x = O(\sqrt{\Delta t})$).

Core Step 2: Taylor Expansion & Score Function Intro

- **Small Perturbation Approximation:** For $\Delta t \rightarrow 0$, $x_{t+\Delta t} \approx x_t$, so perform 1st-order Taylor expansion of $\log p(x_{t+\Delta t})$ at x_t :

$$\log p(x_{t+\Delta t}) \approx \log p(x_t) + \nabla_x \log p(x_t) \cdot \Delta x$$

- **Natural Introduction of Score Function:** Define score function as the gradient of log probability density:

$$\nabla_x \log p(x_t) = \frac{\nabla_x p(x_t)}{p(x_t)} \quad (\text{Score Function})$$

Physical meaning: Points to the direction of fastest density growth.

- **Substitute Expansion into Eq. (7):** Eliminate $\log p(x_{t+\Delta t}) - \log p(x_t)$:

$$\log p(x_t | x_{t+\Delta t}) \approx -\frac{1}{2g^2(t)\Delta t} \|\Delta x - f\Delta t\|^2 - \nabla_x \log p(x_t) \cdot \Delta x$$

Core Step 3: Gaussian Completion via Completing the Square

- Step 3.1: Expand Square Term Ignore high-order term $O(\Delta t^2)$:

$$\|\Delta x - f\Delta t\|^2 \approx \|\Delta x\|^2 - 2\Delta x \cdot f\Delta t$$

Substitute into Eq. (9):

$$\log p(\cdot) \approx -\frac{1}{2g^2\Delta t}(\|\Delta x\|^2 - 2\Delta x \cdot f\Delta t) - \nabla_x \log p \cdot \Delta x$$

- Step 3.2: Rearrange Terms Substitute $\Delta x = x_{t+\Delta t} - x_t$, organize Δx -related terms:

$$\log p(\cdot) \approx -\frac{1}{2g^2\Delta t}\|x_{t+\Delta t} - x_t\|^2 + \left(\frac{f}{g^2} - \nabla_x \log p\right) \cdot (x_{t+\Delta t} - x_t)$$

Core Step 3: Gaussian Completion via Completing the Square

- **Step 3.3: Extract Gaussian Parameters** Compare with $\log \mathcal{N}(x \mid \mu, \Sigma)$:
 - Covariance: $\Sigma_{\text{backward}} = g^2(t) \Delta t \cdot I$ (same as forward);
 - Mean: $\mu_{\text{backward}} = x_{t+\Delta t} + (f - g^2 \nabla_x \log p) \Delta t$;

Core Step 4: Derive Reverse SDE

- **Reverse Process Increment:** Reverse SDE describes $t + \Delta t \rightarrow t$ evolution: $x_t = x_{t+\Delta t} + dx_{\text{backward}}$. The conditional expectation equals Gaussian mean:

$$\mathbb{E}[x_t \mid x_{t+\Delta t}] = \mu_{\text{backward}}$$

- **Drift Term Calculation:** Rearrange Eq. (13) and take $\Delta t \rightarrow 0$ (replace x_t with $x_{t+\Delta t}$):

$$\lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}[x_t - x_{t+\Delta t} \mid x_{t+\Delta t}]}{\Delta t} = f(x, t) - g^2(t) \nabla_x \log p_t(x)$$

- **Noise Term (Time Reversal):** Forward noise: $g(t)dw(t)$; Reverse Brownian motion: $dw_{\text{backward}}(t) = -dw(t)$. Thus reverse noise term: $g(t)dw_{\text{backward}}(t)$.

- **Final Reverse SDE:**

$$dx(t) = (f(x(t), t) - g^2(t) \nabla_x \log p_t(x(t))) dt + g(t) dw_{\text{backward}}(t)$$

Derivation Closure & Key Conclusions

• Rigorous Derivation Pipeline:

- ➊ Forward transfer probability (Gaussian from SDE);
- ➋ Invert via Bayes' theorem to get reverse probability;
- ➌ 1st-order Taylor expansion introduces score function naturally;
- ➍ Complete the square to extract reverse Gaussian parameters;
- ➎ Take $\Delta t \rightarrow 0$ to derive reverse SDE.

• Key Insights:

- **Score Function Role:** Not an artificial assumption - derived from Taylor expansion of log probability difference, embodies gradient of prior density;
- **Drift Correction:** Reverse drift has an extra $-g^2(t)\nabla_x \log p_t(x)$ term, guiding samples to high-density regions;
- **Covariance Consistency:** Reverse diffusion coefficient equals forward one, ensuring probability flow symmetry.